

Review Questions

1. True or False Questions

- (a) If A and B are n by n matrices with $\det(A) = 2$ and $\det(B) = 3$, then $\det(A + B) = 5$.
 - (b) $k\det(A) = \det(kA)$
 - (c) If A is row-equivalent to B , then $\det(A) = \det(B)$.
 - (d) If A is invertible and $AB = 0$, then $B = 0$.
 - (e) If $A^2 = I$, then $A = I$ or $A = -I$.
 - (f) If $A^2 = 0$, then $A = 0$
 - (g) $\det(A) = \det(B)$ implies $A - B$ is not invertible.
 - (h) Let $T(x) = Ax$ and $S(x) = Bx$ and $A \sim B$, then $\text{im}(T) = \text{im}(S)$.
 - (i) Suppose A and B have the same RREF, then the span of the columns of A equals the span of the columns of B .
 - (j) There is a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ so that $\{x : T(x) = 0\} = \{T(x) : x \in \mathbb{R}^2\}$.
 - (k) If u, v, w are pairwise linearly independent, then u, v, w are linearly independent.
2. (a) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the transformation that scales all vectors by a factor of 3. Let A be the standard matrix for T . What is $\det(A)$?
- (b) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the transformation that reflects across the plane $x = z$. Let A be the standard matrix for T . What is $\det(A)$?
- (c) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the transformation that projects to the plane $x = z$. Let A be the standard matrix for T . What is $\det(A)$?

3. Let

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 3 & 3 & 4 \\ 2 & 2 & 3 \end{bmatrix}.$$

Find A^{-1} .

4. (a) Prove that if T is one-to-one and $\vec{u}_1, \dots, \vec{u}_p \in \mathbb{R}^n$ are linearly independent, then $T(\vec{u}_1), \dots, T(\vec{u}_p)$ are linearly independent in $\mathbb{R}^m$.
- (b) In part (a), is the hypothesis that T is one-to-one necessary? Why or why not?
5. Compute the determinant of the following matrices:

(a)

$$A_1 = \begin{bmatrix} 1 & 1 \\ a_1 & 1 \end{bmatrix}.$$

(b)

$$A_2 = \begin{bmatrix} 1 & 1 & 1 \\ a_1 & 1 & 1 \\ a_2 & a_1 & 1 \end{bmatrix}.$$

(c) Find the determinant of the following matrix. Use induction to prove your formula is correct.

$$A_n = \begin{bmatrix} 1 & 1 & \cdots & 1 & 1 \\ a_1 & 1 & \cdots & 1 & 1 \\ \vdots & & & & \vdots \\ a_{n-1} & a_{n-2} & \ddots & 1 & 1 \\ a_n & a_{n-1} & \cdots & a_1 & 1 \end{bmatrix}.$$

6. (a) Prove $|\mathbb{N}| = |\{0\} \cup \mathbb{N}|$.
(b) Prove $|\mathbb{N}| = |\{-2, -1\} \cup \mathbb{N}|$.
(c) Prove $|[a, b]| = |[c, d]|$.

7. Let $T: \mathbb{P}_3 \rightarrow M_{2 \times 2}$ be given by

$$T(a_0 + a_1t + a_2t^2 + a_3t^3) = \begin{bmatrix} a_0 - a_1 & a_2 - a_3 \\ a_1 - a_0 & a_3 - a_2 \end{bmatrix}.$$

Find bases for the kernel and range of L . Justify your answer. Find the standard matrix of L relative to the standard bases.

8. Let $U_{2 \times 2}$ be the space of upper triangular 2 by 2 matrices. Let $T: \mathbb{P}_2 \rightarrow U_{2 \times 2}$ be a linear transformation given by

$$T(f) = \begin{bmatrix} f(0) & f'(1) \\ 0 & f''(2) \end{bmatrix}.$$

Find the matrix of T relative to the standard bases. Find the eigenvalues and eigenspaces of the matrix of T relative to the standard bases.

9. Let T be the linear transformation $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ with matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

- (a) Compute the dimension of the kernel T and a basis for the kernel.
(b) Compute the rank of A and a basis for the image of T .

10. Let $T: \mathbb{P}_2 \rightarrow \mathbb{R}^3$ be defined by

$$T(p(t)) = \begin{bmatrix} p(1) \\ p(2) \\ p(3) \end{bmatrix}$$

for every polynomial $p(t) = a_2t^2 + a_1t + a_0$ in $\mathbb{P}_2$.

- (a) Show that T is a linear transformation.
- (b) Find a matrix M such that $T(p(t)) = M[p(t)]_{\mathcal{B}}$ where $\mathcal{B} = \{1, t, t^2\}$.
- (c) Show that for all $c_1, c_2, c_3 \in \mathbb{R}$ there is a polynomial $p(t) \in \mathbb{P}_2$ such that $p(1) = c_1$, $p(2) = c_2$, and $p(3) = c_3$.

11. Let A be an $n \times n$ matrix which is not invertible, and define K_r to be the nullspace of A^r . That is, $K_1 = \text{Nul}(A)$, $K_2 = \text{Nul}(A^2)$, etc.

- (a) Prove that, for each $j = 1, 2, 3, \dots$, K_j is a subspace of K_{j+1} , so that

$$K_1 \subseteq K_2 \subseteq K_3 \subseteq K_4 \subseteq \dots$$

- (b) Prove that, if for some r we have $K_r = K_{r+1}$, then $K_r = K_n$ for all $n \geq r$.

12. More True or False Questions.

- (a) There is a 2×4 matrix A with rank 1 and

$$\text{Nul}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

- (b) Let V and W be vector spaces and $T : V \rightarrow W$ be a one-to-one linear transformation. If $\{T(v_1), \dots, T(v_k)\}$ is a basis for W , then $\{v_1, \dots, v_k\}$ is a basis for V .
- (c) The set $\{A \in M_{n \times n}(\mathbb{R}) : A \text{ is invertible.}\}$ is a subspace of $M_{n \times n}(\mathbb{R})$.
- (d) Let A be an n by n matrix such that the sum of each row is equal to 1. Then $A - I$ is not invertible.
- (e) Let W_1 and W_2 be 3-dimensional subspaces of $\mathbb{R}^5$, then W_1 and W_2 have a non-zero vector in common.
- (f) V, W subspaces of $\mathbb{R}^n$, then $V \cup W$ is a subspace.
- (g) Let W_1 and W_2 be subspaces of $\mathbb{R}^n$ such that $\dim(W_1) \leq \dim(W_2)$. Then $W_1 \subset W_2$.
- (h) If A is an $n \times n$ matrix and x, y are two eigenvectors of A , then $x + y$ is also an eigenvector of A .
- (i) If V and W are vector spaces of dimension n and $T : V \rightarrow W$ is a one-to-one linear transformation, then T is onto.

Eigenvalues and Eigenvectors – Answers / Solutions

1. (a) **False.** Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$. Then $\det(A + B) = \det \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix} \neq 5$.
 - (b) **False.** Let $k = 2$ and $A = I_2$, then $k \cdot \det A = 2$ and $\det(kA) = 4$.
 - (c) **False.** Let $A = I_2$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then $\det A = 1$ and $\det B = -1$ and A is still row equivalent to B .
 - (d) **True.** Since A is invertible, we can multiply both sides by A^{-1} on the left to get $B = A^{-1}(0) = 0$.
 - (e) **False.** Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then $A^2 = I_2$ and $A \neq I_2$ and $A \neq -I_2$.
 - (f) **False.** Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, then $A^2 = 0$ and $A \neq 0$.
 - (g) **False.** Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$, then $A - B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Thus $\det A = \det B$ and $A - B$ is invertible.
 - (h) **False.** Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, then $A \sim B$. Notice that $\text{im}(T) = \left\{ \begin{bmatrix} x \\ 0 \end{bmatrix} : x \in \mathbb{R} \right\}$ and $\text{im}(S) = \left\{ \begin{bmatrix} 0 \\ y \end{bmatrix} : y \in \mathbb{R} \right\}$. Thus their images are not the same.
 - (i) **False.** Let $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, then A and B have the same reduced row echelon form, but the span of the columns is different.
 - (j) **True.** Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(\vec{x}) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \vec{x}$. Then $\{x : T(x) = 0\} = \text{Span} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$ and $\{T(x) : x \in \mathbb{R}^2\} = \text{im}(T) = \text{Span} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$.
 - (k) **False.** Let $\vec{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and $\vec{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Then each of the three pairs of vectors will be linearly independent, but all three vectors together are linearly dependent.
2. (a) We have that $T(\vec{e}_1) = (3, 0, 0)$, $T(\vec{e}_2) = (0, 3, 0)$ and $T(\vec{e}_3) = (0, 0, 3)$. So the standard matrix for T is $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. Since A is triangular, the determinant of A is the product of the diagonal entries: $\det(A) = 27$.
 - (b) We have that $T(\vec{e}_1) = \vec{e}_3$, $T(\vec{e}_2) = \vec{e}_2$ and $T(\vec{e}_3) = \vec{e}_1$. So the standard matrix for T is $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. Since A is obtained from the identity matrix by exchanging the first and third columns, $\det(A) = -1$.
 - (c) Since T is not onto $\mathbb{R}^3$, T is not invertible. So the standard matrix A for T is not invertible. Thus, $\det(A) = 0$.

3. We augment A by I_3 and row-reduce to find A^{-1} . Thus

$$\begin{aligned} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 3 & 3 & 4 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array} \right) &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 3 & 1 & -3 & 1 & 0 \\ 0 & 2 & 1 & -2 & 0 & 1 \end{array} \right) \\ &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 3 & 1 & -3 & 1 & 0 \\ 0 & 0 & 1 & 0 & -2 & 3 \end{array} \right) \\ &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & -3 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & -2 & 3 \end{array} \right), \end{aligned}$$

and hence $A^{-1} = \begin{bmatrix} 1 & 2 & -3 \\ -1 & 1 & -1 \\ 0 & -2 & 3 \end{bmatrix}$.

4. (a) Consider the vector equation

$$c_1 T(\vec{u}_1) + \cdots + c_n T(\vec{u}_n) = 0. \quad (1)$$

Since T is linear, we have

$$T(c_1 \vec{u}_1 + \cdots + c_n \vec{u}_n) = 0. \quad (2)$$

Since T is one-to-one, we know that if $T(\vec{x}) = 0$, then $\vec{x} = 0$.

Thus we have

$$c_1 \vec{u}_1 + \cdots + c_n \vec{u}_n = 0. \quad (3)$$

Since $\vec{u}_1, \dots, \vec{u}_n$ are linearly independent, we must have $c_1 = c_2 = \cdots = c_n = 0$.

Therefore $T(\vec{u}_1), \dots, T(\vec{u}_n)$ are linearly independent.

(b) The hypothesis that T is one-to-one is necessary. Suppose T is the zero transformation. Let $\vec{u}_1, \dots, \vec{u}_n \in \mathbb{R}^n$ be linearly independent, then $T(\vec{u}_j) = 0$ for all $j = 1, \dots, n$. And hence $T(\vec{u}_1), \dots, T(\vec{u}_n)$ are linearly dependent. Therefore T is a counterexample to the claim without the injectivity hypothesis.

5. (a) In the first case, we get $\det A_1 = 1 - a_1$.

(b) In the second case, we get $\det A_2 = 1 - 2a_1 - a_1^2 = (1 - a_1)^2$.

(c) From the first few cases, it seems like $\det A_n = (1 - a_1)^n$.

Proof. We prove the result by induction.

Base case: We proved the base case in part (a).

Inductive Step: We assume $\det A_n = (1 - a_1)^n$, and we want to prove $\det A_{n+1} = (1 - a_1)^{n+1}$.

We expand the determinant about the first column to get

$$\det A_{n+1} = \det(A_{n+1})_{11} - a_1 \det(A_{n+1})_{21} + \cdots + (-1)^{n+1+1} a_n \det(A_{n+1})_{1(n+1)}.$$

Now notice $(A_{n+1})_{11} = A_n$ and $(A_{n+1})_{21} = A_n$. For $(A_{n+1})_{j1}$, $j > 2$, we have a repeated row of ones, and hence $\det(A_{n+1})_{j1} = 0$ when $j > 2$. Using these facts and the induction hypothesis, we get

$$\det A_{n+1} = \det A_n - a_1 \det A_n = (1 - a_1)^n - a_1(1 - a_1)^n = (1 - a_1)^{n+1}.$$

Thus the result holds by mathematical induction. $\square$

6. (a) We want a bijection from $\mathbb{N}$ to $\{0\} \cup \mathbb{N}$. Define $f : \mathbb{N} \rightarrow \{0\} \cup \mathbb{N}$ by $f(n) = n - 1$. Note f is a bijection.

(b) Define a bijection from $\mathbb{N}$ to $\{-2, -1\} \cup \mathbb{N}$ by $g(k) = \begin{cases} -2 & \text{if } k = 1 \\ -1 & \text{if } k = 2 \\ k - 2 & \text{if } k > 2 \end{cases}$.

(c) Define $h : [a, b] \rightarrow [c, d]$ by $h(x) = c + \frac{d-c}{b-a}(x - a)$.

7. A vector $p(t) = a_0 + a_1t + a_2t^2 + a_3t^3$ will be in the kernel if $a_0 - a_1 = a_2 - a_3 = 0$. Thus a basis for the kernel will be $\{1 + t, t^2 + t^3\}$.

Let $\mathcal{B} = \{1, t, t^2, t^3\}$ be a basis for $\mathbb{P}_3$, and $\mathcal{C} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ be a basis for $M_{2 \times 2}$.

We need to find the matrix $\begin{bmatrix} [T(b_1)]_{\mathcal{C}} & \cdots & [T(b_n)]_{\mathcal{C}} \end{bmatrix}$:

$$\begin{aligned} T(b_1) &= T(1) = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \\ T(b_2) &= T(t) = \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} \\ T(b_3) &= T(t^2) = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \\ T(b_4) &= T(t^3) = \begin{bmatrix} 0 & -1 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Thus the matrix is

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

From the previous part, we know that the range of T is the span of

$$\left\{ \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 0 & 1 \end{bmatrix} \right\}.$$

Removing the dependent entries, we find a basis

$$\mathcal{D} = \left\{ \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \right\}$$

8. Evaluating T on the standard basis for $\mathbb{P}_2$ yields:

$$T(1) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; \quad T(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}; \quad T(t^2) = \begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix}.$$

Let $\mathcal{E}$ denote the standard basis of $U_{2 \times 2}$, then

$$[T(1)]_{\mathcal{E}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \quad [T(t)]_{\mathcal{E}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \quad [T(t^2)]_{\mathcal{E}} = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}.$$

Thus the matrix of T relative to the standard coordinates is given by

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}.$$

We proved in class that the eigenvalues of an upper triangular matrix are given by the diagonal entries, and hence the eigenvalues of T are 1 and 2.

We first find the eigenspace corresponding to the 1-eigenvalue. Thus, we have

$$E_1 = N(M - I) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Similarly, we find the eigenspace for $\lambda = 2$. That is, we have

$$E_2 = N(M - 2I) = \text{Span} \left\{ \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \right\}.$$

Since we have 3 linearly independent eigenvectors, the diagonalization theorem implies T will act diagonally in the bases coming from the eigenvectors. Namely, we have the eigenbasis

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \right\}.$$

This gives the corresponding bases $\mathcal{P} = \{1, t, 2t + t^2\}$ in $\mathbb{P}_2$ and $\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} \right\}$ for $U_{2 \times 2}$. Thus the matrix M of T relative to $\mathcal{P}$ and $\mathcal{B}$ is given by

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

9. (a) The kernel of T is the nullspace of A , so we row reduce A to find

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence a basis for the kernel consists of the one vector

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},$$

which means that the kernel of T is one-dimensional.

- (b) We already row reduced A and we found two pivots, which means that the rank of A is 2.

The image of T is the column space of A and a basis for that space consists of the first two columns of A :

$$\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

10. (a) We must show that $T((p+q)(t)) = T(p(t)) + T(q(t))$ and $T((cp)(t)) = cT(p(t))$ hold for all polynomials $p(t)$ and $q(t)$ in $\mathbb{P}_2$ and every scalar $c \in \mathbb{R}$.

Both equalities are immediate since $(p+q)(t)$ is defined to equal $p(t) + q(t)$ and $(cp)(t)$ is defined to equal $c(p(t))$ for every value of t .

- (b) Write $p(t) = a_2t^2 + a_1t + a_0$ so that $[p(t)]_{\mathcal{B}} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$. Then

$$\begin{bmatrix} p(1) \\ p(2) \\ p(3) \end{bmatrix} = \begin{bmatrix} a_2 + a_1 + a_0 \\ a_24 + a_12 + a_0 \\ a_29 + a_13 + a_0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix},$$

which shows that $M = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}$ works.

- (c) We are asked to solve the equation $T(p(t)) = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$ for the polynomial $p(t) = a_2t^2 + a_1t + a_0$. By part (b), this equation is equivalent to the matrix equation

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}.$$

This equation always has a solution since the matrix M is invertible.

11. (a) We need to show that $K_j \subseteq K_{j+1}$. So suppose $x \in K_j$, that is $A^j x = 0$. Then

$$A^{j+1}x = A(A^j x) = A\mathbf{0} = \mathbf{0},$$

which shows that $x \in K_{j+1}$.

- (b) Suppose that $K_r = K_{r+1}$. We prove the result by induction on $n \geq 1$ that $K_r = K_{r+n}$. The base case $n = 1$ is given to us!

For the induction step, suppose we have shown that $K_r = K_{r+n}$. To show that $K_r = K_{r+n+1}$ we can show instead that $K_{r+n} = K_{r+n+1}$. We already know that $K_{r+n} \subseteq K_{r+n+1}$ so it suffices to show that $K_{r+n+1} \subseteq K_{r+n}$.

Suppose that $x \in K_{r+n+1}$, that is $A^{r+n+1}x = \mathbf{0}$. Then $A^{r+1}(A^n x) = \mathbf{0}$, which means that $A^n x \in K_{r+1}$. Since $K_{r+1} = K_r$, we conclude that $A^n x \in K_r$ and hence $A^r(A^n x) = \mathbf{0}$. Since $A^r A^n = A^{n+r}$, we conclude that $x \in K_{n+r}$.

12. (a) True. An example of such a matrix is

$$A = \begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Indeed, a basis for the null space of this matrix consists of the three vectors

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix},$$

which is also a basis of

$$\text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

since the last spanning vector is redundant.

- (b) True. We need to show that $\{v_1, \dots, v_k\}$ is linearly independent and spans V . Let $v \in V$, then $T(v) \in W$. Since $\{T(v_1), \dots, T(v_k)\}$ is a basis for W , there exists $c_1, \dots, c_k \in \mathbb{R}$ so that

$$T(v) = c_1 T(v_1) + \dots + c_k T(v_k).$$

Since T is injective and linear, we get $v = c_1 v_1 + \dots + c_k v_k$. Thus, $v_1, \dots, v_k$ span V .

We are left to check linear independence. Consider the vector equation $d_1 v_1 + \dots + d_k v_k = 0_V$. Applying T to both sides and using the linearity of T yields

$$d_1 T(v_1) + \dots + d_k T(v_k) = 0.$$

Since $T(v_1), \dots, T(v_k)$ are linearly independent, we know that $d_1 = d_2 = \dots = d_k = 0$. Thus $v_1, \dots, v_k$ are linearly independent, and hence they form a basis.

- (c) False. The zero matrix is not invertible, so the set cannot be a subspace.

- (d) True. Since the sum of each row of A equals 1, we know that 1 is an eigenvalue of A with eigenvector of all 1's. Thus $A - I$ must have non-trivial null space, and hence by the invertible matrix theorem, $A - I_n$ is not invertible.
- (e) True. Let $v \in E_1 \cap E_2$. Let E_j be the eigenspace corresponding to λ_j for $j = 1, 2$. Then $\lambda_1 v = \lambda_2 v$ since $\lambda_1 \neq \lambda_2$, we must have $v = 0$.
- (f) True. Let x_1, x_2, x_3 be a basis for W_1 and y_1, y_2, y_3 be a basis for W_2 . Since $x_1, x_2, x_3, y_1, y_2, y_3$ is 6 vectors in $\mathbb{R}^5$, the vectors must be linearly dependent. Thus there are scalars, not all zero, such that

$$c_1x_1 + c_2x_2 + c_3x_3 + c_4y_1 + c_5y_2 + c_6y_3 = 0,$$

and hence

$$c_1x_1 + c_2x_2 + c_3x_3 = -c_4y_1 - c_5y_2 - c_6y_3.$$

Thus $c_1x_1 + c_2x_2 + c_3x_3$ is a non-zero element in $W_1 \cap W_2$.

- (g) False. Let $V = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ and $W = \text{Span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$. Now $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \in V \cup W$, but $\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin V \cup W$.
- (h) False. Let $W_1 = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ and $W_2 = \text{Span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$. Then $\dim W_1 \leq \dim W_2$, but W_1 is not a subset of W_2 .
- (i) True. Because T is one-to-one, the dimension of the image of T must be n . Since W has dimension n , the image of T must equal W .